

Statement of Purpose

Program: Master of Engineering (ME) in Computational Science and Engineering

Applicant: Yang Sen (Liam) Lin

In my recent research accepted to **CoRL 2025**, titled “*See, Point, Fly*,” I successfully achieved state-of-the-art zero-shot UAV navigation by geometrically decoupling vision from control. While this modular approach was effective, I encountered a critical computational deadlock: geometric rules are inherently brittle in dynamic, unstructured environments, yet the alternative—end-to-end Generative Visual-Language-Action (VLA) models—is currently too computationally heavy for real-time edge control. This experience clarified that the future of Physical AI requires satisfying two opposing constraints: the **agile adaptability** of generative reasoning and the **extreme efficiency** of embedded systems.

I am applying to the **Master of Engineering (ME) in Computational Science and Engineering** at Harvard to bridge this gap. My specific research objective is to investigate **generative control policies** that are mathematically robust yet algorithmically optimized for **resource-constrained deployment**. I aim to achieve this by utilizing the 2-year thesis timeline to integrate **Prof. Yilun Du’s** compositional generative frameworks with **Prof. Vijay Janapa Reddi’s** edge computing expertise.

My journey toward this “computational bridge” began with a focus on algorithmic rigor. For my CoRL work, I derived a geometric action-to-control framework to map 2D VLM prompts to 3D metric controls. Leveraging my background in **Automatic Control (I) & (II) (A+/A)**, I designed a closed-loop **Adaptive Step System** inspired by PID principles that solved depth ambiguity without depth sensors. Mathematically, I optimized the visual tracking complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$ via heuristic sparse sampling, enabling the system to run on standard hardware. However, despite these optimizations, the system relies on deterministic geometric priors. It fails when the environment requires non-linear reasoning, such as dodging erratic obstacles. To enable true autonomy, I must move from deterministic geometry to **Stochastic Generative Policies**, which requires the deep theoretical foundation in Energy-Based Models offered by the CSE curriculum.

Simultaneously, I recognized that advanced algorithms are useless if they saturate onboard hardware. During my internship at **Wolley Inc.**, I engineered C-based diagnostic tools to profile **Memory Bandwidth Pressure** for CXL server interconnects. While CXL is a server-scale technology, the fundamental bottleneck—**managing memory bandwidth under high-throughput constraints**—is exactly what plagues VLM deployment on embedded drones. This experience gave me the low-level architectural intuition needed to optimize I/O efficiency for physical AI. I plan to apply this optimization mindset during my upcoming internship at **OMRON SINIC X**, where I will distill Robotics Foundation Models for soft robotics. Yet, I realize that software distillation alone is insufficient; deploying foundation models on drones requires a rigorous investigation into hardware-software co-design, a core focus of the CSE program.

The **Harvard CSE ME** is the only program that combines rigorous mathematical modeling with high-performance computing, offering the ideal environment to merge these distinct research tracks over a dedicated thesis. The curriculum is the rigorous toolkit I need. My strong performance in **Engineering**

Mathematics (I) & (II) (A-/A-), **Fluid Mechanics (A+)**, and **Heat Transfer (A+)** has grounded me in the differential equation rigor required for **AM 205 (Numerical Methods)**. Furthermore, **AM 207 (Stochastic Methods)** and **CS 2050 (High Performance Computing)** provide the necessary probabilistic and systems foundations to transition my research from geometric to generative models.

Research-wise, I seek to work with **Prof. Yilun Du** to upgrade my geometric approach to a **Generative Policy** approach, exploring how Energy-Based Models can capture complex dynamics. Simultaneously, I aim to collaborate with **Prof. Vijay Janapa Reddi** to **investigate the trade-offs** of applying TinyML quantization to these diffusion policies, determining if we can retain expressivity while meeting the millisecond-latency requirements of a drone. The **Master of Engineering thesis** is the necessary vehicle to integrate these disparate fields—Stochastic Algorithms and Edge Systems—over a dedicated 2-year timeline.

My long-term goal is to lead R&D in Physical AI, solving the scalability of autonomous agents. Harvard CSE will transform me from a Robotics Engineer who builds systems into a **Computational Scientist** capable of inventing the mathematical frameworks that define the next generation of autonomy.